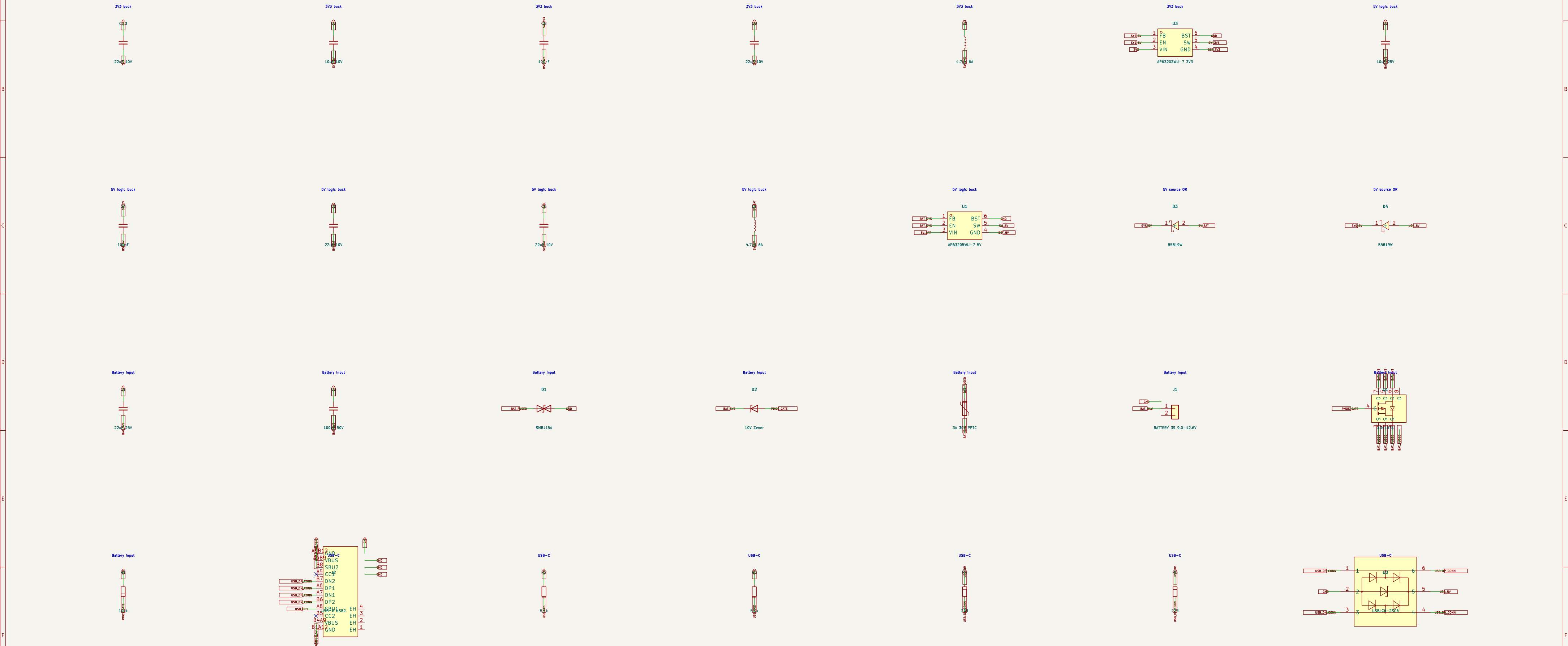


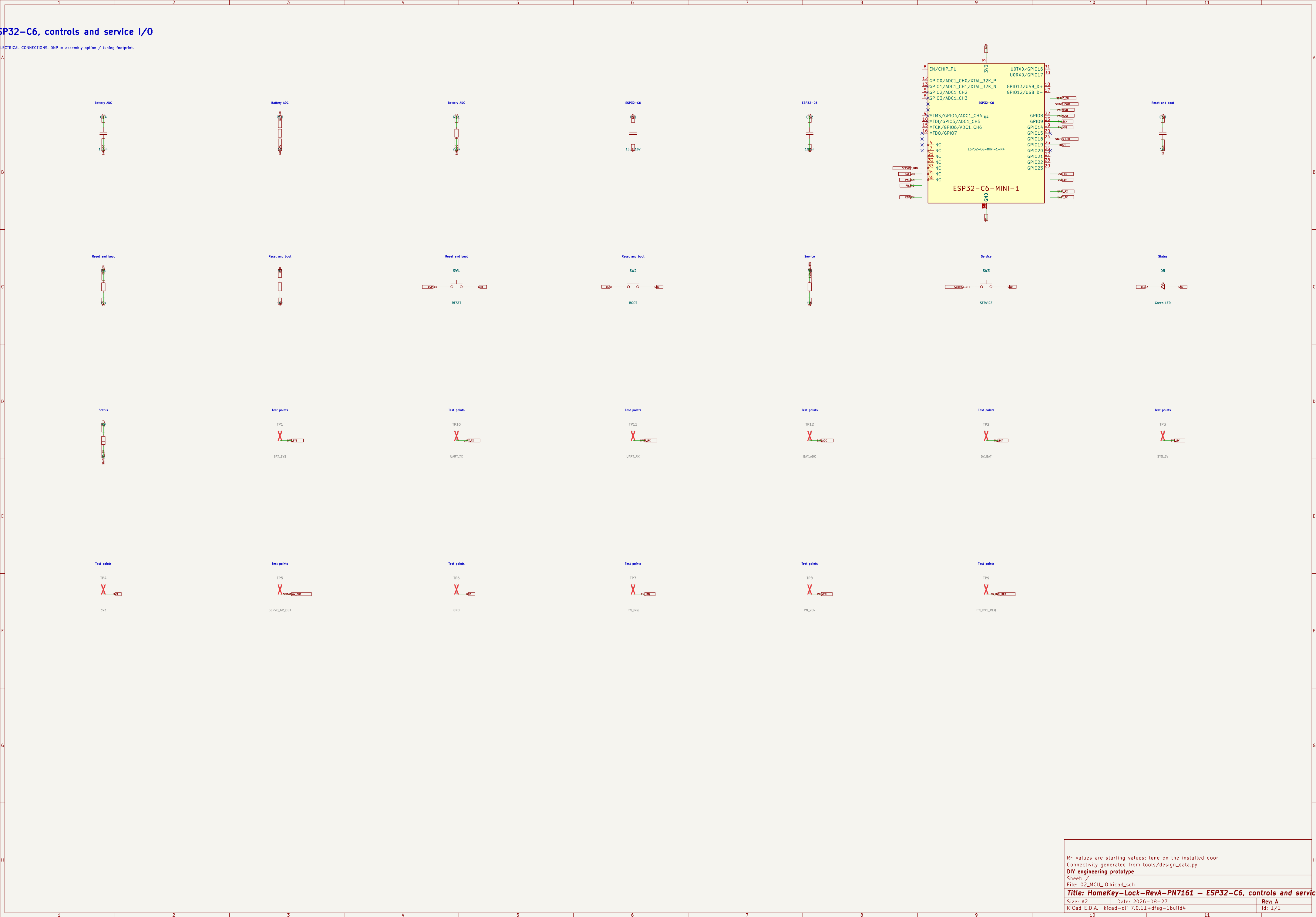
B — Power input, USB and always-on rails

VALUES ARE THE ELECTRICAL CONNECTIONS. DNP = assembly option / tuning footprint.



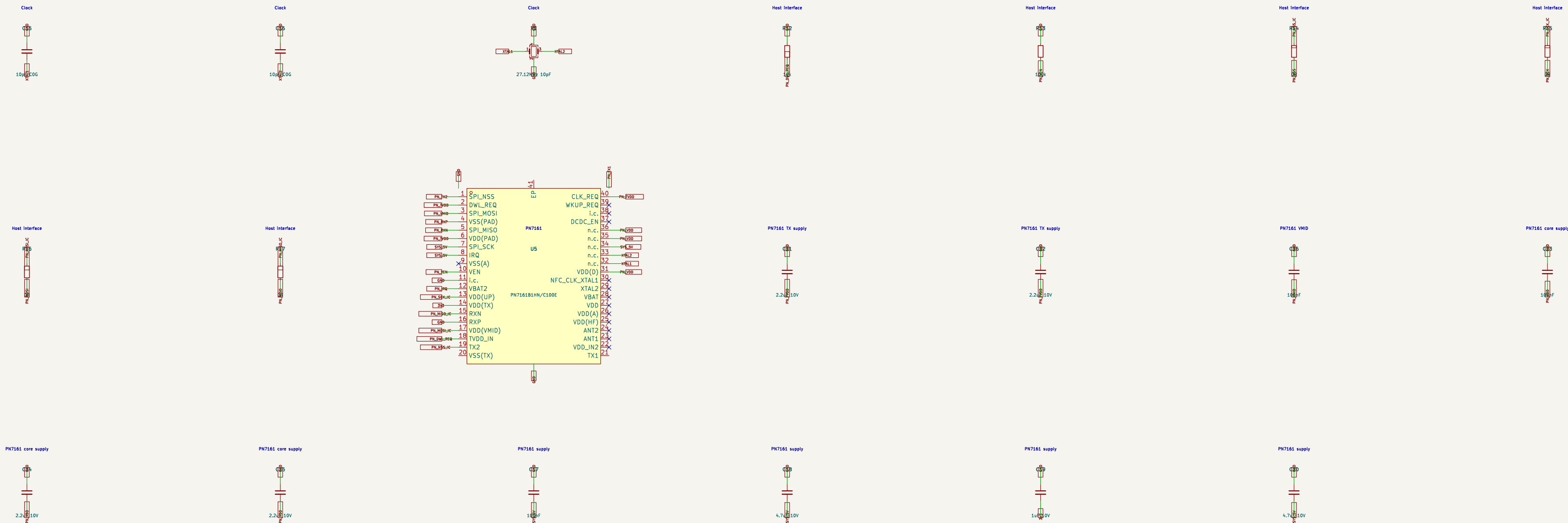
– ESP32–C6, controls and service I/O

ARE THE ELECTRICAL CONNECTIONS. DNP = assembly option / tuning footprint.



CONTROLLER — PN7161 controller, clock and supplies

ARE THE ELECTRICAL CONNECTIONS, DNP = assembly option / tuning footprint.



RF values are starting values; tune on the installed door
Connectivity generated from tools/design_data.py

DIY engineering prototype

Sheet: /
File: 03_NFC_CONTROLLER.kicad_sch

Title: HomeKey-Lock-RevA-PN7161 – PN7161 controller, clock and supplies

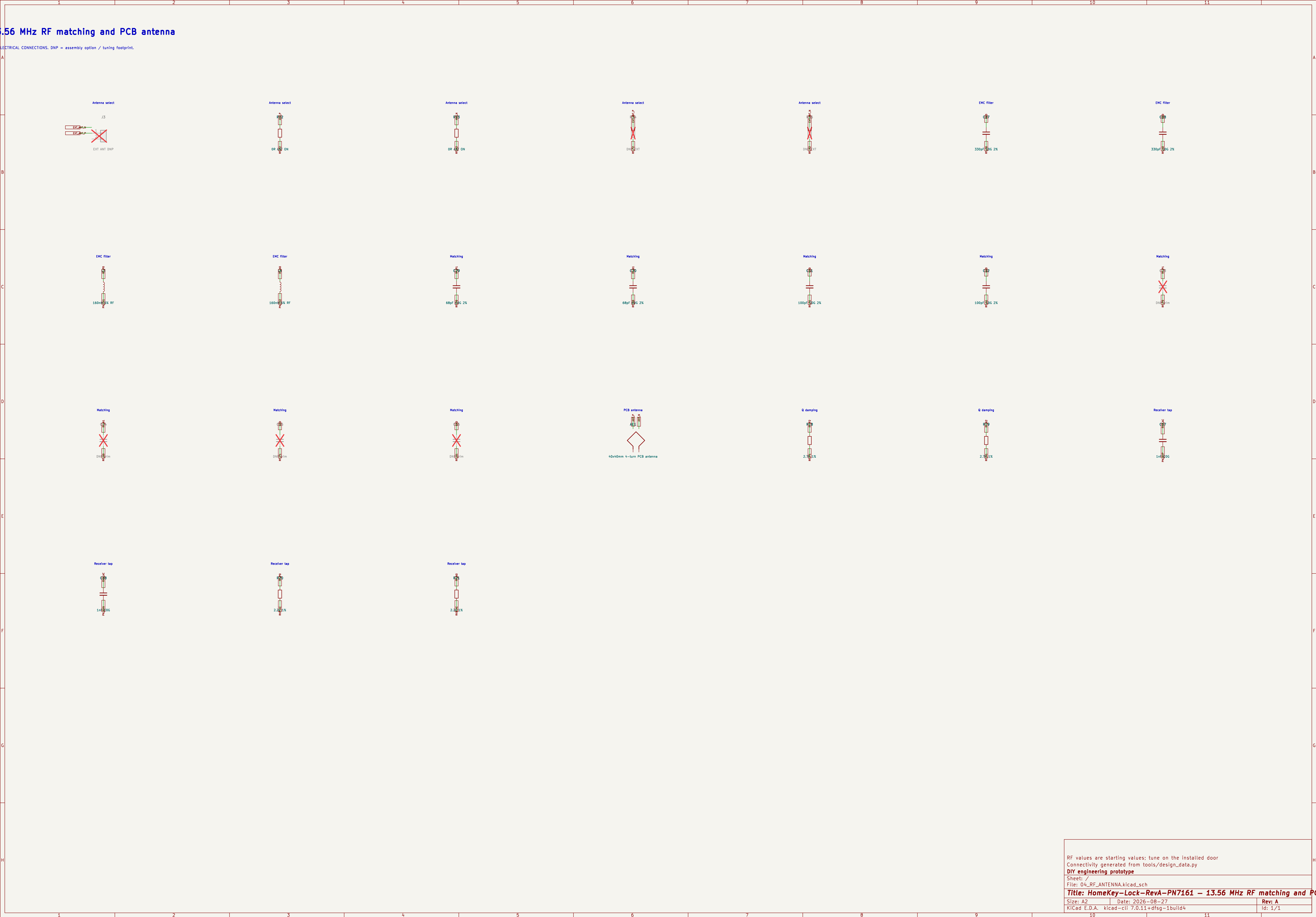
Size: A2 Date: 2026-08-27

KiCad E.D.A. kicad-cli 7.0.11+dfsg-1

Rev: A

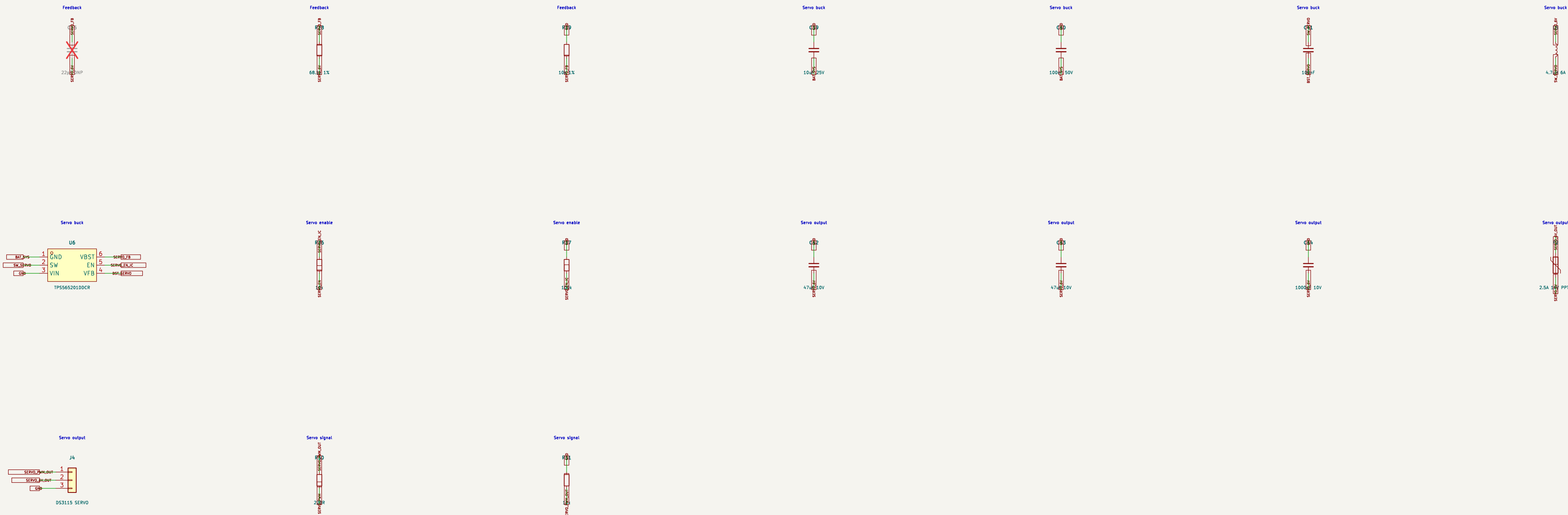
— 13.56 MHz RF matching and PCB antenna

ARE THE ELECTRICAL CONNECTIONS. DNP = assembly option / tuning footprint.



VER – Switched 5.98 V / 5 A servo supply

ARE THE ELECTRICAL CONNECTIONS. DNP = assembly option / tuning footprint.



RF values are starting values; tune on the installed door
Connectivity generated from tools/design_data.py

DIY engineering prototype

Title: HomeKey-Lock-RevA-PN7161 – Switched 5.98 V / 5 A servo supply

Size: A2	Date: 2026-08-27	Rev: A
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1